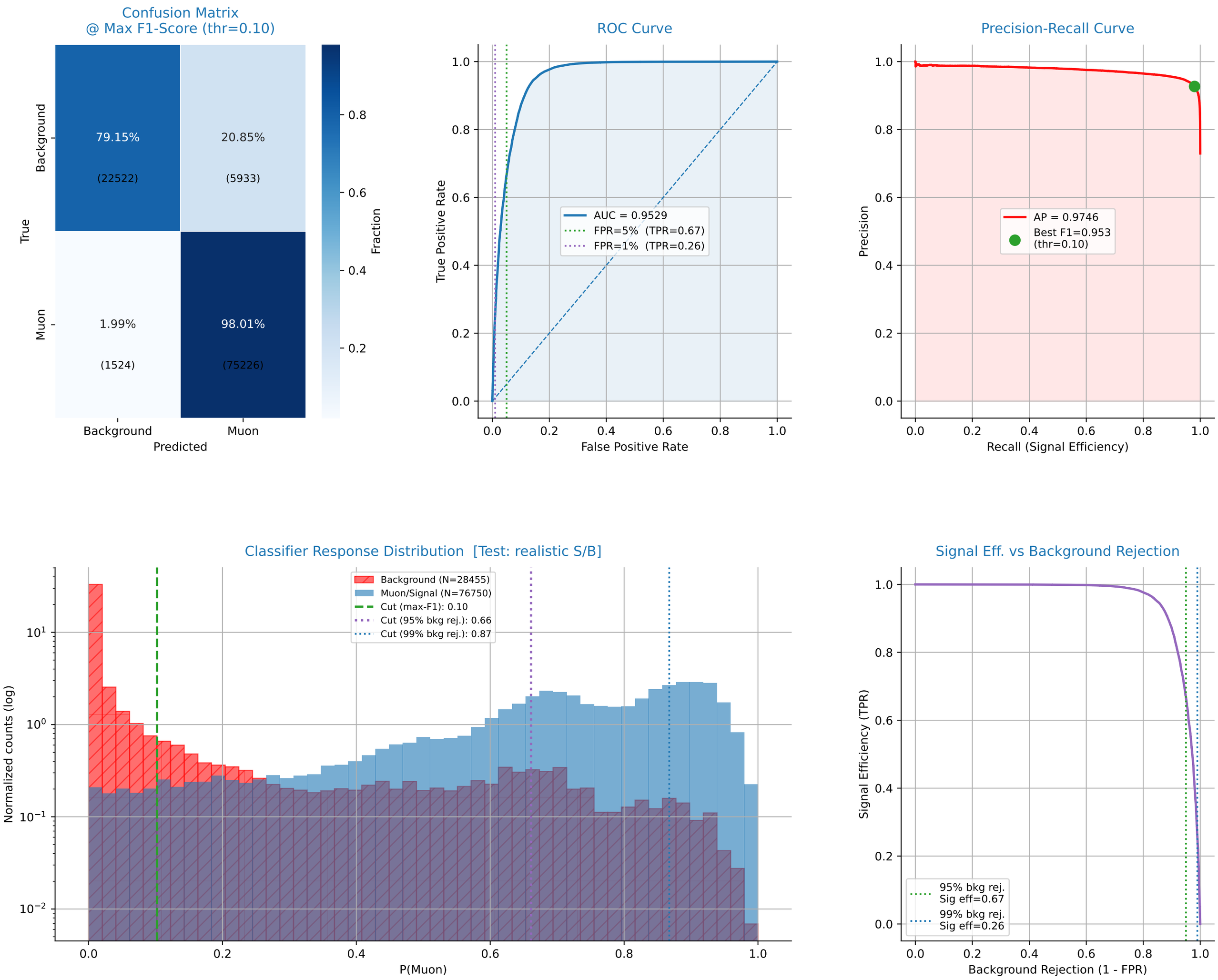
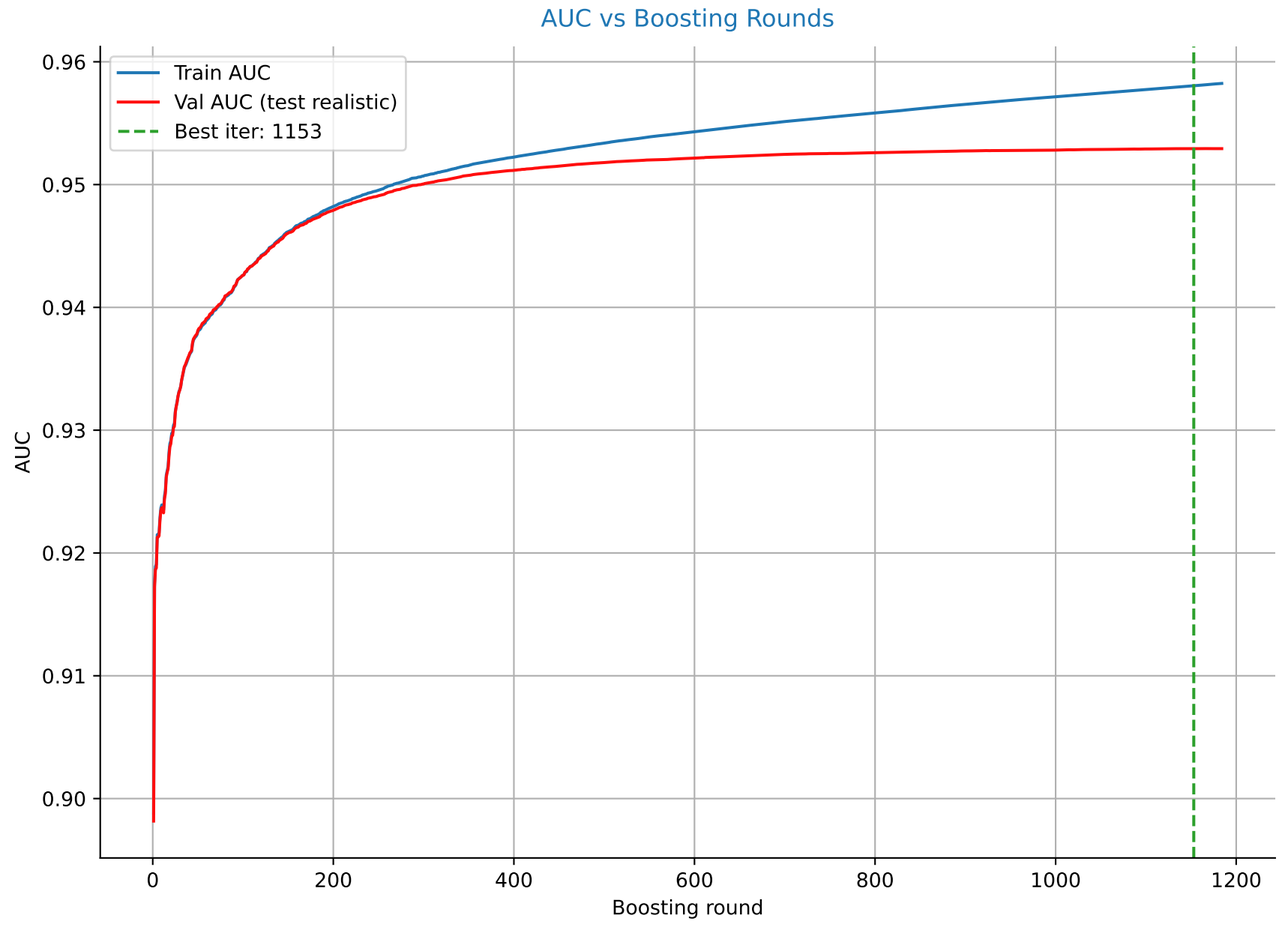
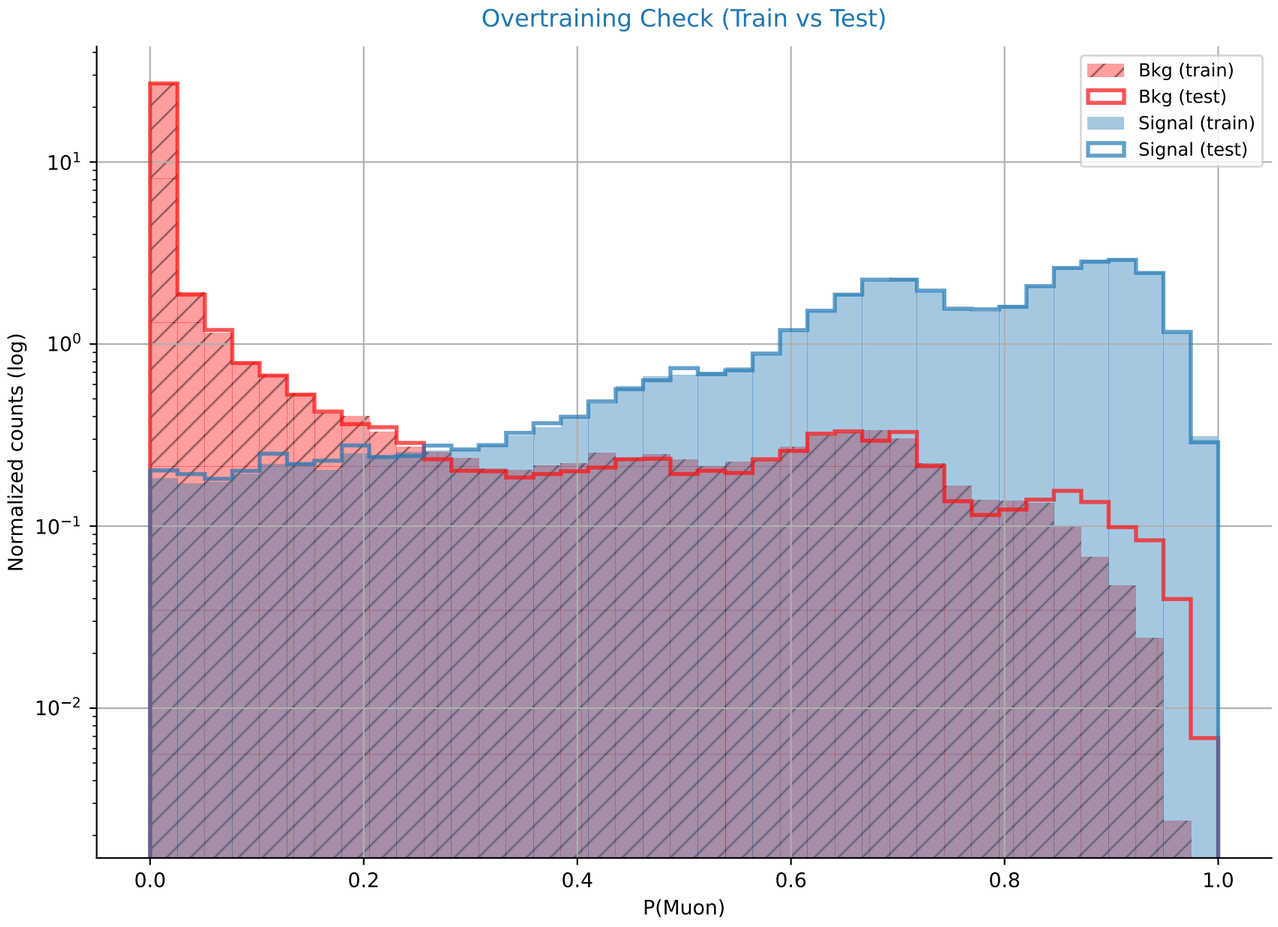
Muon Identification — XGBoost Analysis
[Train S/B=1:0 | Test S/B=1:0 (realistic, all files)]

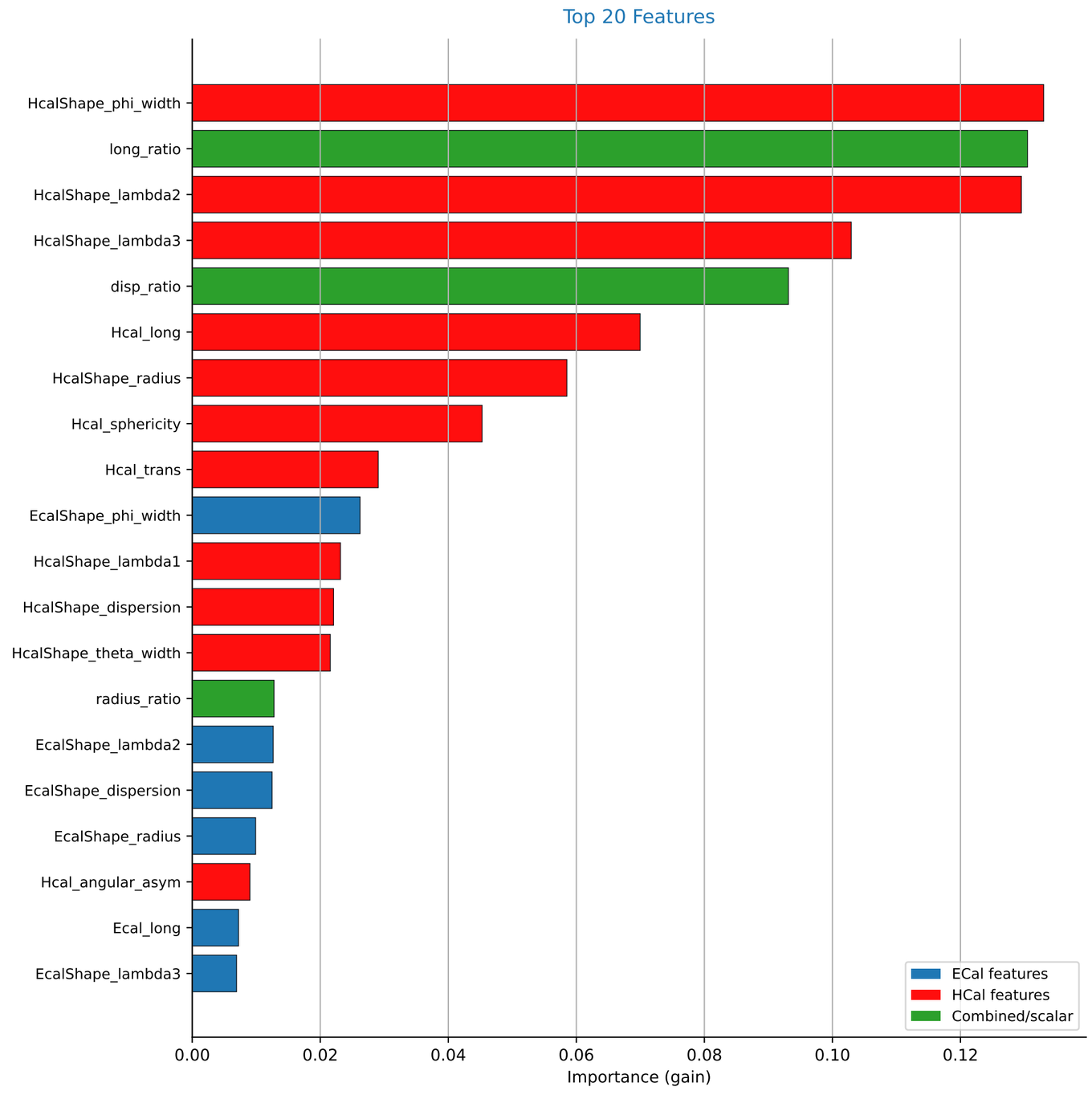


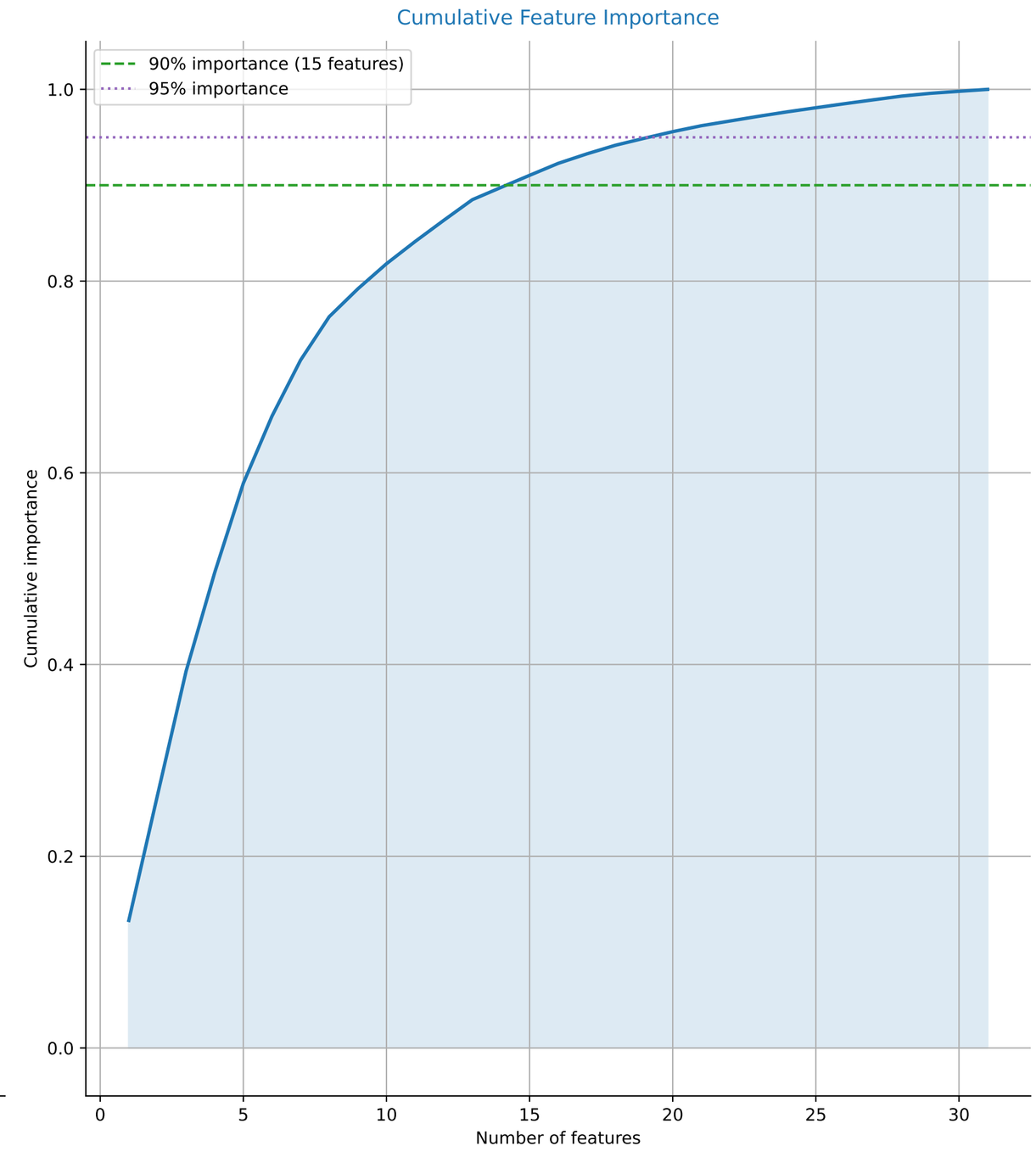
XGBoost Training Diagnostics



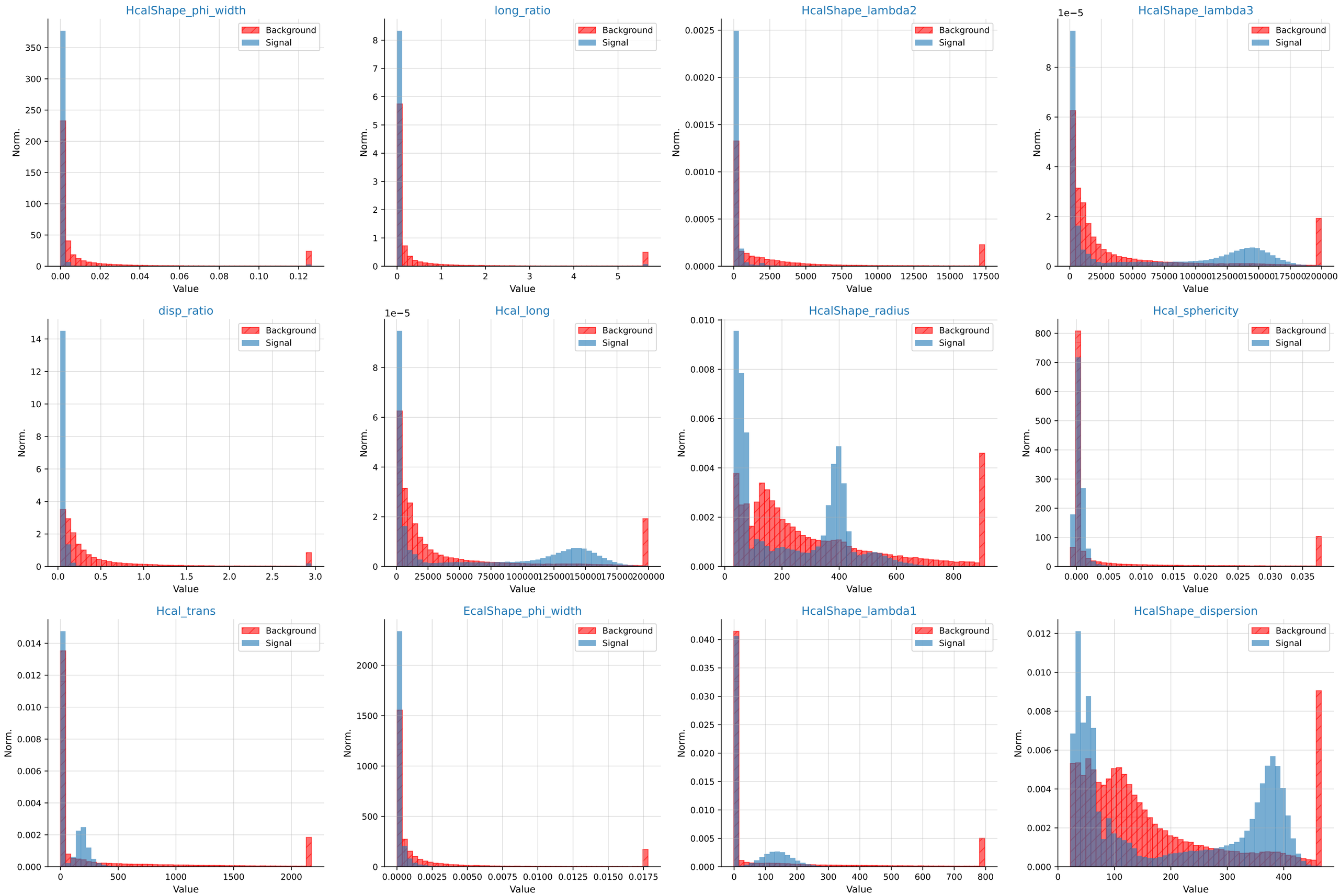


Feature Importances — XGBoost



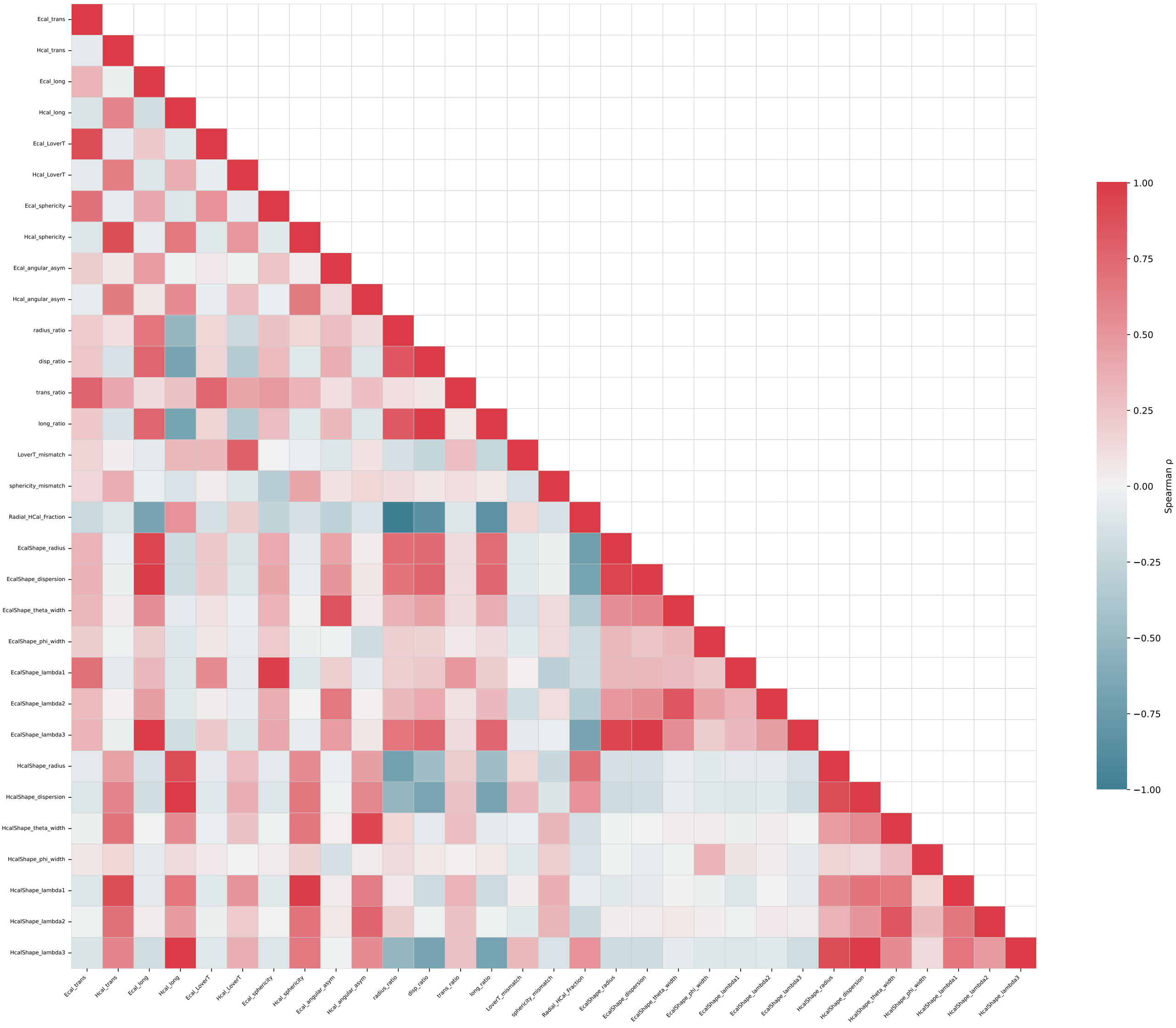


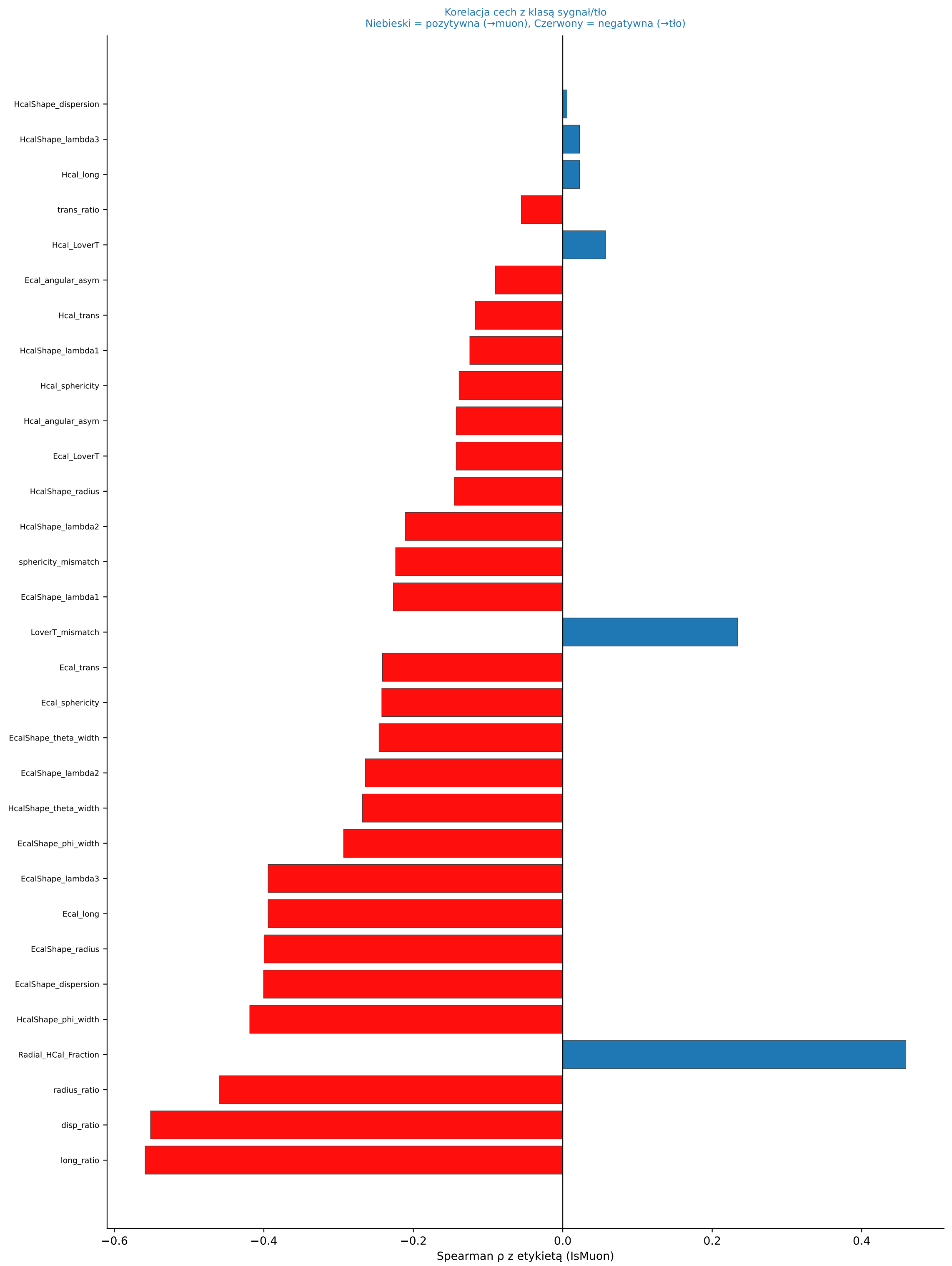
Key Physics Variable Distributions: Signal vs Background
[rozkłady z całego datasetu przed podziałem]

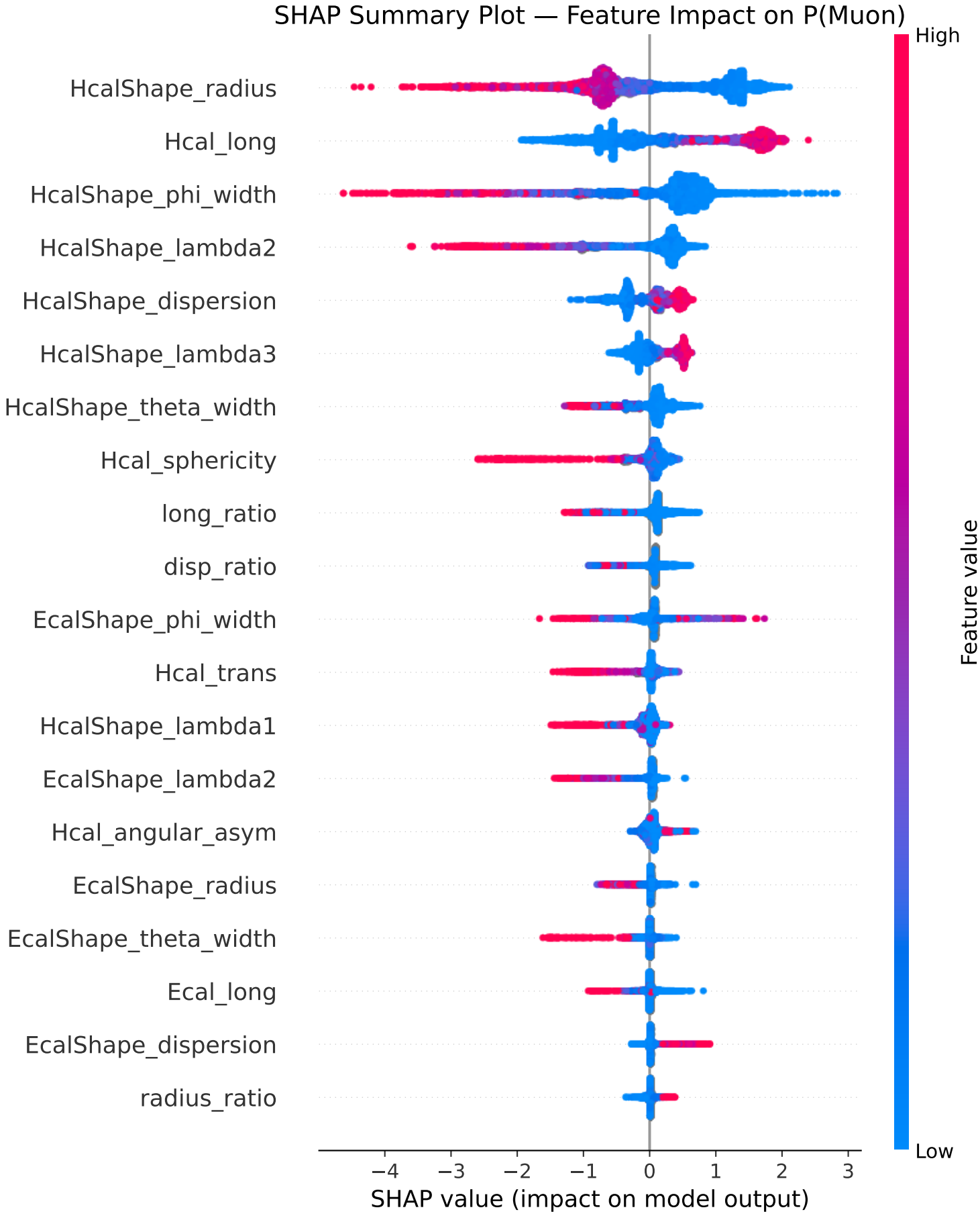


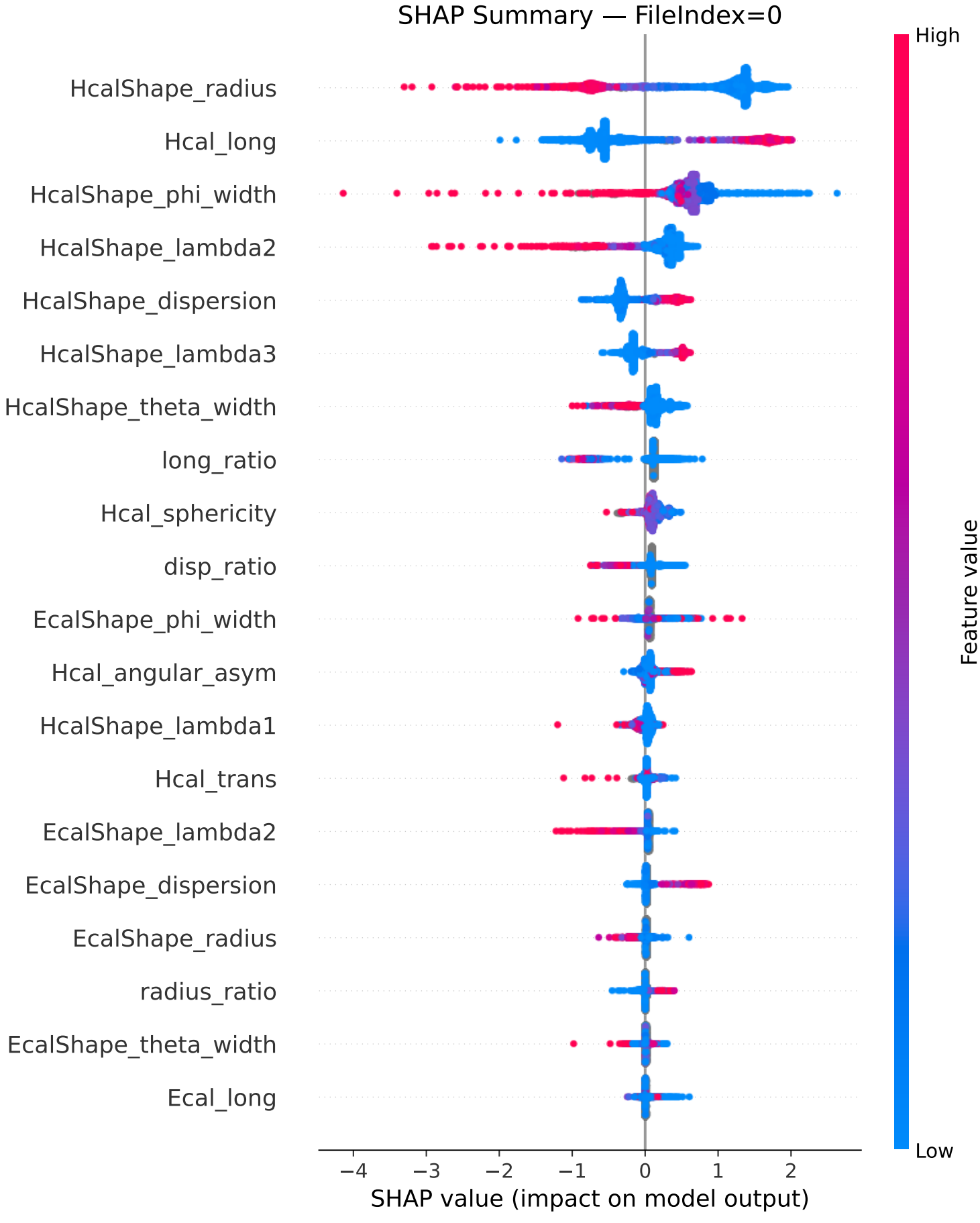
Spearman Correlation Matrix — All Features
(wartości z całego datasetu)

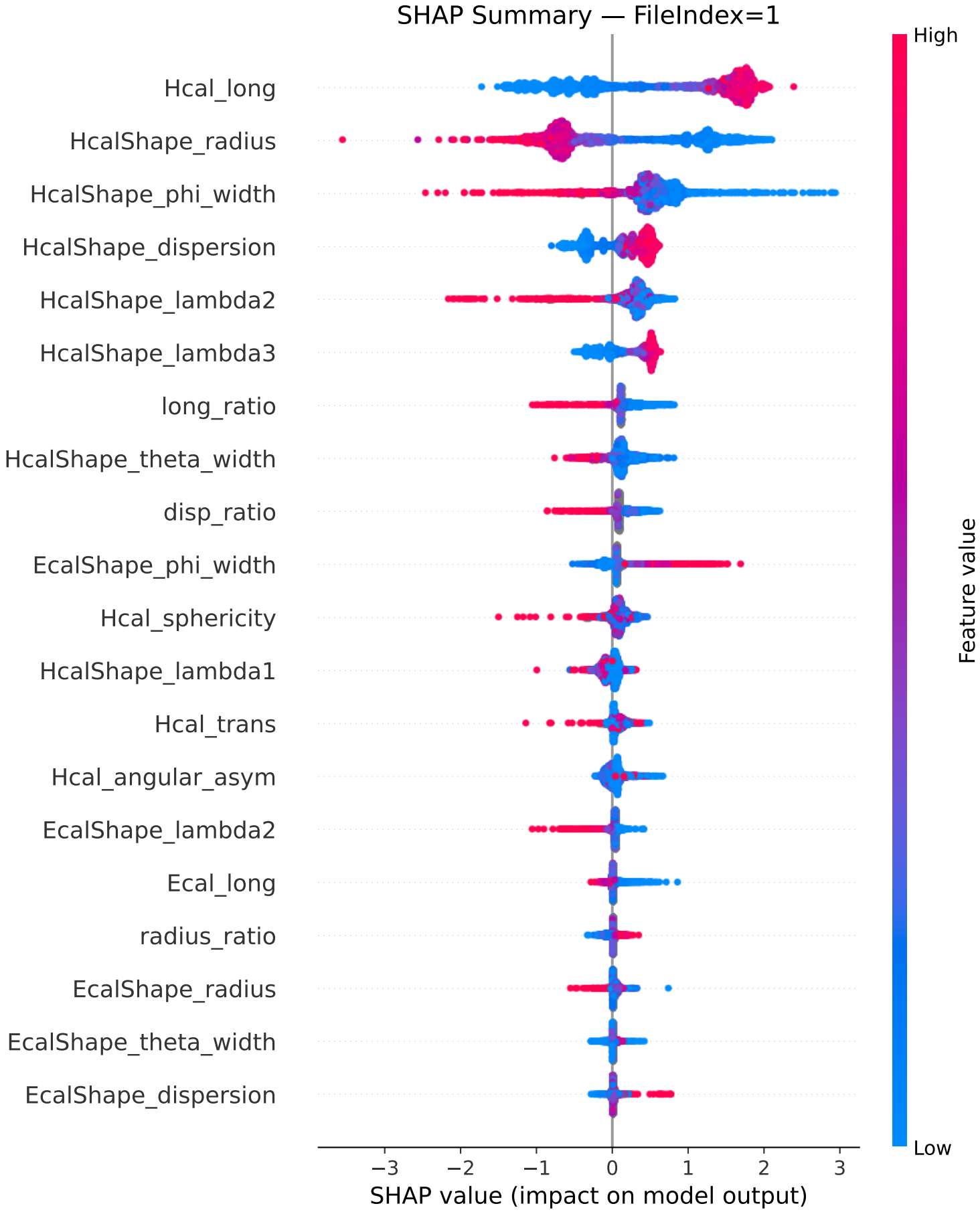
Wyższe |ρ| → silniejsza monotonna zależność między cechami

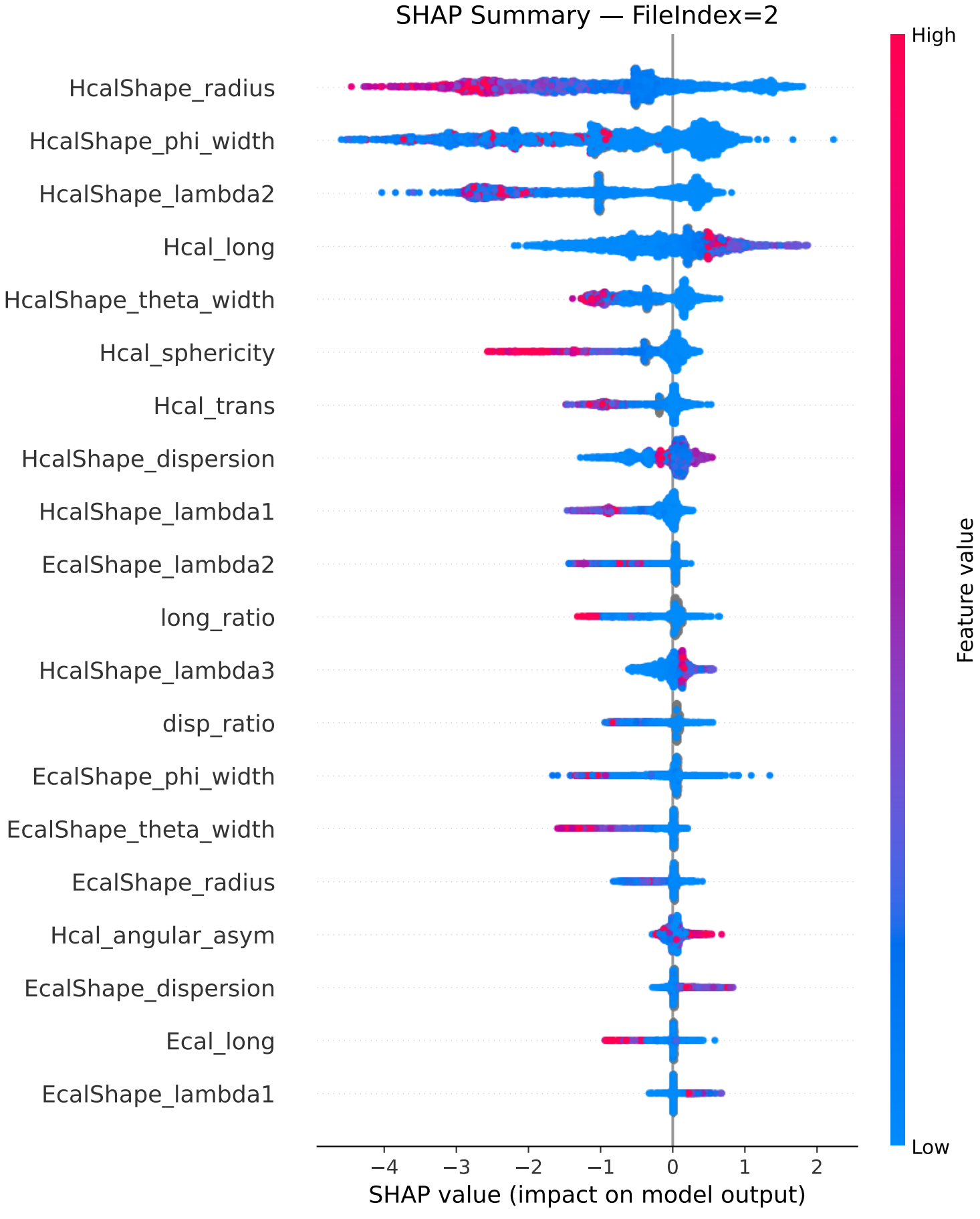


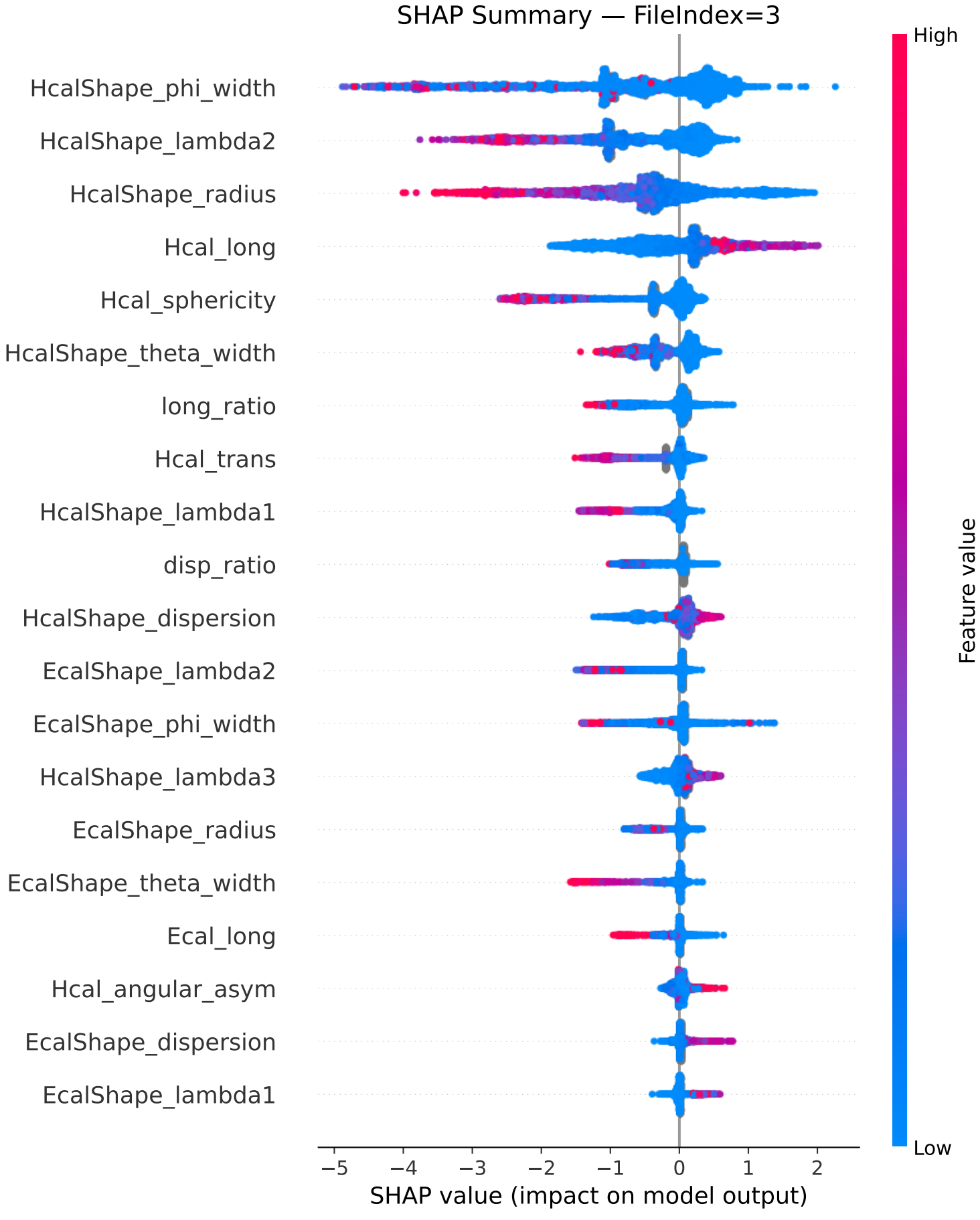

Korelacja cech z klasą sygnał/tło
Niebieski = pozytywna (→muon), Czerwony = negatywna (→tło)
HcalShape_dispersion
HcalShape_lambda3
Hcal_long
trans_ratio
Hcal_LoverT
Ecal_angular_asym
Hcal_trans
HcalShape_lambda1
Hcal_sphericity
Hcal_angular_asym
Ecal_LoverT
HcalShape_radius
HcalShape_lambda2
sphericity_mismatch
EcalShape_lambda1
LoverT_mismatch
Ecal_trans
Ecal_sphericity
EcalShape_theta_width
EcalShape_lambda2
HcalShape_theta_width
EcalShape_phi_width
EcalShape_lambda3
Ecal_long
EcalShape_radius
EcalShape_dispersion
HcalShape_phi_width
Radial_HCal_Fraction
radius_ratio
disp_ratio
long_ratio
−0.6
−0.4
−0.2
0.0
0.2
0.4
Spearman ρ z etykietą (IsMuon)

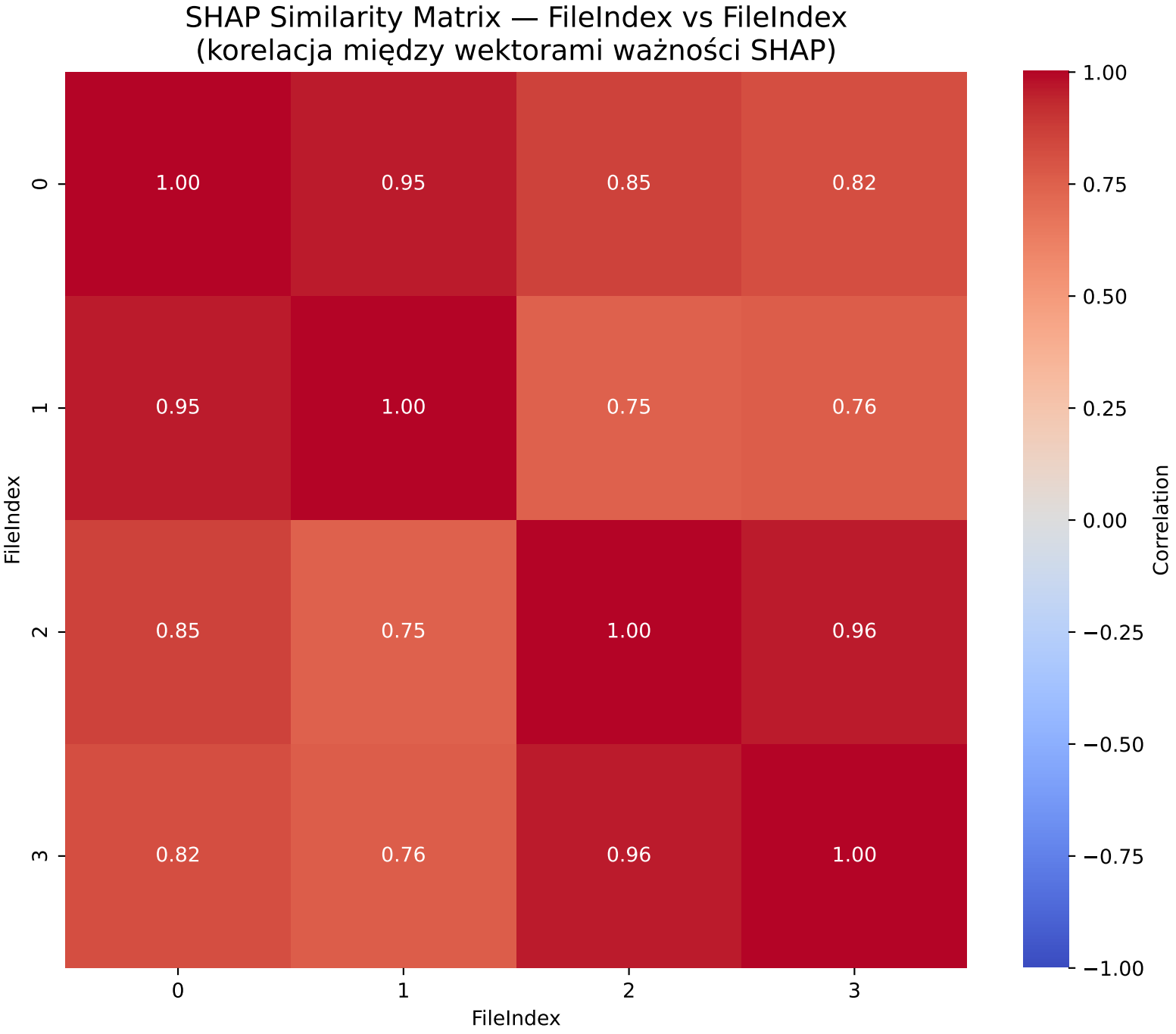

SHAP Summary Plot — Feature Impact on P(Muon)
HcalShape_radius
Hcal_long
HcalShape_phi_width
HcalShape_lambda2
HcalShape_dispersion
HcalShape_lambda3
HcalShape_theta_width
Hcal_sphericity
long_ratio
disp_ratio
EcalShape_phi_width
Hcal_trans
HcalShape_lambda1
EcalShape_lambda2
Hcal_angular_asym
EcalShape_radius
EcalShape_theta_width
Ecal_long
EcalShape_dispersion
radius_ratio
−4
−3
−2
−1
0
1
2
3
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=0
HcalShape_radius
Hcal_long
HcalShape_phi_width
HcalShape_lambda2
HcalShape_dispersion
HcalShape_lambda3
HcalShape_theta_width
long_ratio
Hcal_sphericity
disp_ratio
EcalShape_phi_width
Hcal_angular_asym
HcalShape_lambda1
Hcal_trans
EcalShape_lambda2
EcalShape_dispersion
EcalShape_radius
radius_ratio
EcalShape_theta_width
Ecal_long
−4
−3
−2
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=1
Hcal_long
HcalShape_radius
HcalShape_phi_width
HcalShape_dispersion
HcalShape_lambda2
HcalShape_lambda3
long_ratio
HcalShape_theta_width
disp_ratio
EcalShape_phi_width
Hcal_sphericity
HcalShape_lambda1
Hcal_trans
Hcal_angular_asym
EcalShape_lambda2
Ecal_long
radius_ratio
EcalShape_radius
EcalShape_theta_width
EcalShape_dispersion
−3
−2
−1
0
1
2
3
SHAP value (impact on model output)
High
Low
Feature value


SHAP Summary — FileIndex=2
HcalShape_radius
HcalShape_phi_width
HcalShape_lambda2
Hcal_long
HcalShape_theta_width
Hcal_sphericity
Hcal_trans
HcalShape_dispersion
HcalShape_lambda1
EcalShape_lambda2
long_ratio
HcalShape_lambda3
disp_ratio
EcalShape_phi_width
EcalShape_theta_width
EcalShape_radius
Hcal_angular_asym
EcalShape_dispersion
Ecal_long
EcalShape_lambda1
−4
−3
−2
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=3
HcalShape_phi_width
HcalShape_lambda2
HcalShape_radius
Hcal_long
Hcal_sphericity
HcalShape_theta_width
long_ratio
Hcal_trans
HcalShape_lambda1
disp_ratio
HcalShape_dispersion
EcalShape_lambda2
EcalShape_phi_width
HcalShape_lambda3
EcalShape_radius
EcalShape_theta_width
Ecal_long
Hcal_angular_asym
EcalShape_dispersion
EcalShape_lambda1
−5
−4
−3
−2
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low


SHAP Similarity Matrix — FileIndex vs FileIndex
(korelacja między wektorami ważności SHAP)
1.00
0.95
0.85
0.82
0.95
1.00
0.75
0.76
0.85
0.75
1.00
0.96
0.82
0.76
0.96
1.00
FileIndex
0
1
2
3
FileIndex
0
1
2
3
Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00